Week 2

Lecture 3 Subspaces

This covers part of ***Section 1.2*** *in the course notes.*

Recall from MATH 136 that in $\mathbb{R}^n$, we can define lines and planes. A line in $\mathbb{R}^n$ is formed by taking two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ with $\vec{v} \neq \vec{0}$ and defining:

$$L = \{\vec{u} + t\vec{v} : t \in \mathbb{R}\}.$$

This is the unique line which goes through $\vec{u}$ in the direction $\vec{v}$.

Similarly a plane is formed by taking three vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ with $\vec{v}, \vec{w} \neq 0$ and $\vec{v} \neq c\vec{w}$ for all $c \in \mathbb{R}$, and defining:

$$P = \{\vec{u} + s\vec{v} + t\vec{w} : s, t \in \mathbb{R}\}.$$

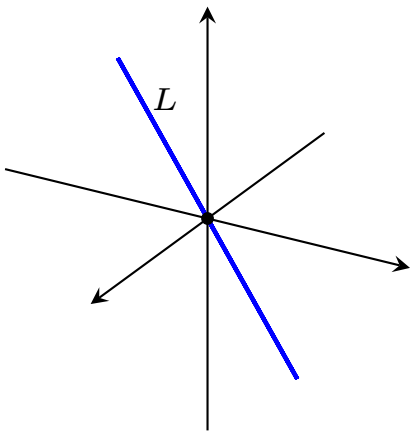
This is the unique line which goes through $\vec{u}$ in the directions $\vec{v}$ and $\vec{w}$.

We can say more about lines and planes if we assume that $\vec{u} = \vec{0}$ (that is, if the lines and planes we define go through the origin). For a line, this would mean:

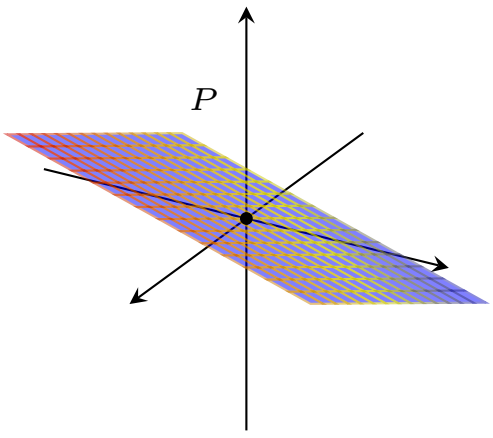
$$L = \{t\vec{v} : t \in \mathbb{R}\}$$

and for a plane:

$$P = \{s\vec{v} + t\vec{w} : s, t \in \mathbb{R}\}.$$



A line L in $\mathbb{R}^3$



A plane P in $\mathbb{R}^3$

In this case, it turns out L and P are each vector spaces by themselves under the operations of addition and scalar multiplication inherited from $\mathbb{R}^n$.

Our goal now is to generalize this idea as follows: given an abstract vector space V, we want to

- define an analogue for the notion of a vector space lying inside another vector space, and
- determine an easy way to check if a given subset of V is itself a vector space.

A vector space which lies inside another one has a special name, one you should recognize from MATH 136.

Definition 3.1 (Subspace)

Let V be a vector space over $\mathbb{F}$. A **subspace** of V is a subset $U \subseteq V$ which is itself a vector space under the addition and scalar multiplication operations inherited from V.

Example 3.2

If $0 \leq m \leq n$ are integers, then $\mathcal{P}_m(\mathbb{F})$ is a subspace of $\mathcal{P}_n(\mathbb{F})$. This is because $\mathcal{P}_m(\mathbb{F}) \subseteq \mathcal{P}_n(\mathbb{F})$ and we already know that $\mathcal{P}_m(\mathbb{F})$ is itself a vector space under the operations inherited from $\mathcal{P}_n(\mathbb{F})$.

Also $\mathcal{P}_n(\mathbb{F})$ is a subspace of $\mathcal{P}(\mathbb{F})$ for any $n \geq 0$.

Example 3.3 ($\{\vec{0}\}$ and V are subspaces of V)

Within *any* vector space V over $\mathbb{F}$, we have the zero vector space $\{\vec{0}\}$ (where the zero $\vec{0}$ is the same zero coming from V) and the whole space V are subspaces of V.

This is because $\{\vec{0}\}$ is itself a vector space (as we saw, this is simply the zero vector space) and any vector space V is tautologically a vector space.

There are also other examples of subspaces which are less obvious to see than the ones above. For such examples, we invoke the following result which gives us an easy way to check:

Proposition 3.4 (Subspace test)

Let V be a vector space over $\mathbb{F}$ and let $U \subseteq V$ be a subset. Then U is a subspace of V if and only if all three of the following properties hold:

(i) U is non-empty.

(ii) U is closed under addition.

(iii) U is closed under scalar multiplication.

Proof.

($\implies$) Suppose first that U is a subspace. Then by **[A2]**, U has some zero vector $\vec{0}_U \in U$, so U is non-empty. U must also closed under addition and scalar multiplication by virtue of indeed being a vector space. Thus, **(i)**, **(ii)**, and **(iii)** all hold.

Notice also that if $\vec{0}_V$ is the zero vector of V, then $\vec{0}_U = 0 \cdot \vec{0}_U = \vec{0}_V$, so the zero vectors are in fact the same and we can just write it as $\vec{0}$.

($\impliedby$) Now suppose $U \subseteq V$ is a subset which satisfies all three properties listed above. By **(ii)**, the addition operation on V when restricted only to the vectors in U indeed maps into U, so $+ : U \times U \to U$ makes sense. Similarly by **(iii)**, the same is true for scalar multiplication so $\cdot : \mathbb{F} \times U \to U$ makes sense.

Now we can verify all of the vector space properties hold. All of these hold trivially from the fact that they hold for V, except for the following two:

[A2] Because of **(i)**, we can find some $\vec{u} \in U$. Then by **(iii)**, we have $0\vec{u} = \vec{0} \in U$.

[A3] Given any $\vec{u} \in U$, by **(iii)** we have $(-1)\vec{u} = -\vec{u} \in U$.

This completes our proof. □

Often it is useful to combine **(ii)** and **(iii)** above:

Corollary 3.5 (Subspace test – concise version)

Let V be a vector space over $\mathbb{F}$ and let $U \subseteq V$ be a subset. Then U is a subspace of V if and only if

(i) U is non-empty, and

(ii) for all $\vec{u}_1, \vec{u}_2 \in U$ and all $\alpha \in \mathbb{F}$, we have $\alpha\vec{u}_1 + \vec{u}_2 \in U$.

Proof. Exercise. □

Example 3.6

Let $\vec{v}, \vec{w} \in \mathbb{R}^n$ be non-zero vectors with $\vec{v} \neq c\vec{w}$ for all $c \in \mathbb{R}$ (notice this last condition forces $n \geq 2$).

We consider the line L and plane P in $\mathbb{R}^n$ given by:

$$L = \{t\vec{v} : t \in \mathbb{R}\},$$
$$P = \{s\vec{v} + t\vec{w} : s, t \in \mathbb{R}\}.$$

To check L is a subspace, we can apply the subspace test:

(i) We know $\vec{0} = 0\vec{v} \in L$.

(ii) If $\vec{x}, \vec{y} \in L$ and $\alpha \in \mathbb{R}$, we have $\vec{x} = t_1\vec{v}$ and $\vec{y} = t_2\vec{v}$ for some $t_1, t_2 \in \mathbb{R}$. Thus:
$$\alpha\vec{x} + \vec{y} = \alpha t_1\vec{v} + t_2\vec{v} = (\alpha t_1 + t_2)\vec{v} \in L.$$

A similar application of the subspace test on P shows that this too is a subspace of $\mathbb{R}^n$.

Let's now look at some other examples.

Example 3.7

Consider the vector space $\mathcal{P}(\mathbb{F})$ of polynomials with coefficients in $\mathbb{F}$. Let

$$U = \{p(x) \in \mathcal{P}(\mathbb{F}) : p(2) = 0\}.$$

We can apply the subspace test to verify that U is a subspace of $\mathcal{P}(\mathbb{F})$:

(i) Notice that the zero polynomial 0 is in U, so U is non-empty.

(ii) If $p(x), q(x) \in U$ and $\alpha \in \mathbb{F}$, then for $f(x) = \alpha p(x) + q(x)$ we have:

$$f(2) = \alpha p(2) + q(2) = \alpha \cdot 0 + 0 = 0$$

which shows $f(x) = \alpha p(x) + q(x) \in U$.

Example 3.8 (Upper triangular matrices)

Consider the vector space $\mathcal{M}_n(\mathbb{F}) = \mathcal{M}_{n\times n}(\mathbb{F})$ over $\mathbb{F}$ of $n \times n$ matrices with entries in $\mathbb{F}$. Define:

$$\mathcal{U}_n(\mathbb{F}) = \{A = [a_{i,j}] \in \mathcal{M}_n(\mathbb{F}) : a_{i,j} = 0 \text{ for all } i > j\}.$$

A matrix in $\mathcal{U}_n(\mathbb{F})$ is called **upper triangular**. This means that every entry on the lower triangle of a matrix in $\mathcal{U}_n(\mathbb{F})$ is zero. That is, each $A = [a_{i,j}] \in \mathcal{U}_n(\mathbb{F})$ has the form:

$$A = [a_{i,j}] = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n} \end{bmatrix}.$$

An easy application of the subspace test shows that $\mathcal{U}_n(\mathbb{F})$ is a subspace of $\mathcal{M}_n(\mathbb{F})$:

(i) Certainly it is non-empty since $O_{n\times n} \in \mathcal{U}_n(\mathbb{F})$.

(ii) If $A = [a_{i,j}]$ and $B = [b_{i,j}]$ are in $\mathcal{U}_n(\mathbb{F})$, and $\alpha \in \mathbb{F}$, then $\alpha A + B = [\alpha a_{i,j} + b_{i,j}]$ still satisfies that $\alpha a_{i,j} + b_{i,j} = \alpha \cdot 0 + 0 = 0$ for any $i > j$, so $\alpha A + B \in \mathcal{U}_n(\mathbb{F})$.

Example 3.9

Consider the vector space $\mathcal{C}([0,1])$ of all continuous functions from $[0,1]$ to $\mathbb{R}$. Let:

$$\mathcal{C}^1([0,1]) = \{f \in \mathcal{C}([0,1]) : f \text{ is differentiable and } f' \text{ is continuous}\}$$

where $f' = \frac{df}{dx}$ is the derivative of f. We see that this set is non-empty since for example 0 is inside it. Also, for any $f, g \in \mathcal{C}^1([0,1])$ and any $\alpha \in \mathbb{R}$, $\alpha f + g$ is differentiable with $(\alpha f + g)' = \alpha f' + g'$, which is also continuous. Thus, $\mathcal{C}^1([0,1])$ is a subspace of $\mathcal{C}([0,1])$.

Exercise 3.10

Consider again the vector space $\mathcal{P}(\mathbb{F})$. Define:

$$U = \{0\} \cup \{p(x) \in \mathcal{P}(\mathbb{F}) : \deg(p(x)) \text{ is even}\}$$

$$W = \{0\} \cup \{p(x) \in \mathcal{P}(\mathbb{F}) : \deg(p(x)) \text{ is odd}\}$$

Determine whether or not U and W are subspaces of $\mathcal{P}(\mathbb{F})$.

Exercise 3.11 (Diagonal matrices)

Consider again the vector space $\mathcal{M}_n(\mathbb{F}) = \mathcal{M}_{n\times n}(\mathbb{F})$ over $\mathbb{F}$. Define:

$$\mathcal{D}_n(\mathbb{F}) = \{A = [a_{i,j}] \in \mathcal{M}_n(\mathbb{F}) : a_{i,j} = 0 \text{ for all } i \neq j\}.$$

A matrix in $\mathcal{D}_n(\mathbb{F})$ is called **diagonal**, which means every entry which is not on the diagonal of a matrix in $\mathcal{D}_n(\mathbb{F})$ is zero.

Show that $\mathcal{D}_n(\mathbb{F})$ is a subspace of $\mathcal{M}_n(\mathbb{F})$. Notice in particular that $\mathcal{D}_n(\mathbb{F})$ is a subspace of $\mathcal{U}_n(\mathbb{F})$ too since $\mathcal{D}_n(\mathbb{F}) \subseteq \mathcal{U}_n(\mathbb{F})$.

Lecture 4 Span and linear independence

This covers part of ***Section 1.2*** *and part of* ***Subsection 1.3.1*** *in the course notes.*

We now look at the notions of *span* and *independence*. In the next lecture, we shall see that these are in fact related, and together yield a way in which we can make sense of a "coordinate system" in certain vector spaces.

The notion of span gives us a way to create subspaces of a given vector space out of a set of vectors. Remember that when we created a line in $\mathbb{R}^n$ going through the origin, we took a single non-zero vector $\vec{u} \in \mathbb{R}^n$ and defined:

$$L = \{t\vec{u} : t \in \mathbb{R}\}$$

and when we created a plane through the origin, we took two non-zero vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ with the property that $\vec{u} \neq c\vec{v}$ for all $c \in \mathbb{R}$ and defined:

$$P = \{s\vec{u} + t\vec{v} : s, t \in \mathbb{R}\}.$$

In each example, we created a subspace out of one or two direction vectors. The idea for abstract vector spaces is exactly the same. Let us fix a vector space V over $\mathbb{F}$.

Definition 4.1 (Span of a set of vectors)

Let $S = \{\vec{v}_1, ..., \vec{v}_n\}$ be a finite subset V. The **span** of S is given by:

$$\text{Span}(S) = \{\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n : \alpha_i \in \mathbb{F}\}.$$

We will also define $\text{Span}(\emptyset) = \{\vec{0}\}$. The elements of $\text{Span}(S)$ are called **linear combinations** of the vectors $\vec{v}_1, ..., \vec{v}_n$.

Remark 4.2

We can also define the span of any subset $S \subseteq V$ (S could be infinite) as follows:

$$\text{Span}(S) = \{\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n : \alpha_i \in \mathbb{F}, \vec{v}_i \in S\}$$

if $S \neq \emptyset$, and $\text{Span}(\emptyset) = \{\vec{0}\}$ (can you see why this new definition of span agrees with the one we defined earlier if S is finite?) The elements of $\text{Span}(S)$ are also called **linear combinations** of the elements of S.

Since we will largely be studying *finite-dimensional* linear algebra in this course, we will not need to worry about this definition.

With spans, we can form lines and planes more concisely:

Example 4.3

To form a line in $\mathbb{R}^n$ going through the origin, we take non-zero $\vec{v} \in \mathbb{R}^n$ and set:

$$L = \text{Span}(\{\vec{v}\}).$$

To form a plane in $\mathbb{R}^n$ going through the origin, we take non-zero $\vec{v}, \vec{w} \in \mathbb{R}^n$ satisfying $\vec{v} \notin \text{Span}(\{\vec{w}\})$ and set:

$$P = \text{Span}(\{\vec{v}, \vec{w}\}).$$

Let's check that in the setting of an abstract vector space, we can be sure the span is indeed a subspace.

Proposition 4.4

For any subset $S = \{\vec{v}_1, ..., \vec{v}_n\} \subseteq V$, $\text{Span}(S)$ is a subspace of V.

Moreover, it is in fact the *smallest* subspace of V which contains all the vectors $\vec{v}_1, ..., \vec{v}_n$ as elements.

Proof. If $S = \emptyset$ (the case where $n = 0$), we see that $\text{Span}(S) = \{\vec{0}\}$ is just the zero subspace.

So assume $n \geq 1$ so that $S \neq \emptyset$. Since $\vec{0} \in \text{Span}(S)$, certainly $\text{Span}(S)$ is non-empty. Now let $\vec{u}, \vec{v} \in \text{Span}(S)$ and $\alpha \in \mathbb{F}$. Then we can write:

$$\vec{u} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n,$$

$$\vec{v} = \beta_1\vec{v}_1 + \cdots + \beta_n\vec{v}_n$$

for some $\alpha_i, \beta_i \in \mathbb{F}$. Now notice:

$$\vec{u} + \vec{v} = (\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n) + (\beta_1\vec{v}_1 + \cdots + \beta_n\vec{v}_n) = (\alpha_1 + \beta_1)\vec{v}_1 + \cdots + (\alpha_n + \beta_n)\vec{v}_n$$

and this is in $\text{Span}(S)$ by definition. Hence, we have closure under addition. Similarly:

$$\alpha\vec{u} = \alpha(\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n) = (\alpha\alpha_1)\vec{v}_1 + \cdots + (\alpha\alpha_n)\vec{v}_n$$

is also in $\text{Span}(S)$, so we have closure under scalar multiplication too. Thus, $\text{Span}(S)$ is indeed a subspace of V.

Now let U be any subspace of V which contains all the vectors in S. We want to prove then that $\text{Span}(S) \subseteq U$.

But indeed, anything in $\text{Span}(S)$ is of the form $\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n$ for some $\alpha_i \in \mathbb{F}$. Since U contains S, we have each $\vec{v}_i \in U$. By the properties of closure under addition and scalar multiplication of a subspace, it follows that $\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n \in U$ as well, which means $\text{Span}(S) \subseteq U$ as claimed. □

In the case where $\text{Span}(S) = V$, we have a special name for S.

Definition 4.5 (Spanning set for V)

Let $S = \{\vec{v}_1, ..., \vec{v}_n\} \subseteq V$. We say S is a **spanning set for V** when $\text{Span}(S) = V$.

Example 4.6

Consider the subspace $\mathcal{P}_n(\mathbb{F})$ of $\mathcal{P}(\mathbb{F})$. Any polynomial $p(x)$ of degree $\leq n$ is of the form $p(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_n x^n$ for $\alpha_i \in \mathbb{F}$, and vice versa. Hence, it is easy to see that $\mathcal{P}_n(\mathbb{F}) = \text{Span}(\{1, x, x^2, ..., x^n\})$. In particular, $\{1, x, x^2, ..., x^n\}$ is a spanning set for $\mathcal{P}_n(\mathbb{F})$.

There are cases where a different choice of spanning vectors will give us the same subspace. For example, we know from the above that $\mathcal{P}_2(\mathbb{F}) = \text{Span}\{1, x, x^2\}$. But consider the set $S = \{1, 1 + x, x + x^2\}$ and let $U = \text{Span}(S)$. Since any linear combination of S is a polynomial of degree ≤ 2, we have $U \subseteq \mathcal{P}_2(\mathbb{F})$. Conversely though, suppose $\alpha_0 + \alpha_1 x + \alpha_2 x^2 \in \mathcal{P}_2(\mathbb{F})$ is given. Then we can write:

$$\begin{aligned}
\alpha_0 + \alpha_1 x + \alpha_2 x^2 &= \alpha_0 + \alpha_1(1 + x - 1) + \alpha_2(x + x^2 - x) \\
&= (\alpha_0 - \alpha_1) + \alpha_1(1 + x) + \alpha_2(x + x^2) - \alpha_2 x \\
&= (\alpha_0 - \alpha_1) + \alpha_1(1 + x) + \alpha_2(x + x^2) - \alpha_2(1 + x - 1) \\
&= (\alpha_0 - \alpha_1 + \alpha_2) + (\alpha_1 - \alpha_2)(1 + x) + \alpha_2(x + x^2)
\end{aligned}$$

which shows $\mathcal{P}_2(\mathbb{F}) \subseteq U$ as well, hence $U = \mathcal{P}_2(\mathbb{F})$.

Later, we will see more efficient ways we can prove the equality of two spanning sets when the examples get more complicated.

Exercise 4.7

Consider the vector space $\mathcal{M}_n(\mathbb{F})$ of $n \times n$ matrices over $\mathbb{F}$ and the subspaces $\mathcal{U}_n(\mathbb{F})$ and $\mathcal{D}_n(\mathbb{F})$ as defined in **Example 3.8** and **Exercise 3.11**, respectively.

Find matrices $A_1, ..., A_k \in \mathcal{M}_n(\mathbb{F})$ and $B_1, ..., B_l \in \mathcal{M}_n(\mathbb{F})$ for which $\{A_1, ..., A_k\}$ forms a spanning set for $\mathcal{U}_n(\mathbb{F})$ and $\{B_1, ..., B_l\}$ forms a spanning set for $\mathcal{D}_n(\mathbb{F})$. What do you think is the smallest number k you need to span $\mathcal{U}_n(\mathbb{F})$? What about the smallest l to span $\mathcal{D}_n(\mathbb{F})$?

We now turn to the second notion of this lecture: *independence*.

Definition 4.8

Let $S = \{\vec{v}_1, ..., \vec{v}_n\}$. We say S is **linearly independent** when the following holds:

$$\text{for all } \alpha_1, ..., \alpha_n \in \mathbb{F}, \text{ if } \alpha_1 \vec{v}_1 + \cdots + \alpha_n \vec{v}_n = \vec{0} \text{ then } \alpha_1 = \cdots = \alpha_n = 0.$$

In other words, the only solution to the equation $\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n = \vec{0}$ is $(\alpha_1, ..., \alpha_n) = (0, ..., 0)$.

If S is not linearly independent, we say it is **linearly dependent**. We also declare the empty set $\emptyset$ to be linearly *independent*.

Example 4.9

In $\mathcal{P}_2(\mathbb{F})$, the set $\{1, x, x^2\}$ is linearly independent. For, if $\alpha_0+\alpha_1 x+\alpha_2 x^2 = 0$, then we must have every $\alpha_i = 0$ (otherwise the left-hand side would be a non-zero polynomial equal to the zero polynomial!)

On the other hand, the set $\{1, 3x, 1-2x, 4x^2-2x+1\}$ is linearly *dependent*. Indeed, notice that the equation:

$$\alpha_0 \cdot 1 + \alpha_1 \cdot (3x) + \alpha_2 \cdot (1-2x) + \alpha_3 \cdot (4x^2 - 2x + 1) = 0$$

has a non-trivial solution: $(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = (1, -\frac{2}{3}, -1, 0)$ (please check!)

A useful way of checking if a set (with two or more elements) is linearly dependent is given by the following

Proposition 4.10

Let $S = \{\vec{v}_1, ..., \vec{v}_n\} \subseteq V$ with $n \geq 2$. Then S is linearly dependent if and only if one of the $\vec{v}_i$'s can be written as a linear combination of the other ones in S.

Proof. If we have linear dependence, then we can find $\alpha_1, ..., \alpha_n \in \mathbb{F}$ *not all zero* such that:

$$\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n = \vec{0}$$

By reordering the vectors in S if necessary, we can assume without loss of generality that $\alpha_1 \neq 0$. Then we have:

$$\vec{v}_1 = -\frac{1}{\alpha}(\alpha_2\vec{v}_2 + \cdots + \alpha_n\vec{v}_n)$$

and this is a linear combination of $\{\vec{v}_2, ..., \vec{v}_n\}$ (this set is non-empty since $n \geq 2$).

Conversely, suppose one of the $\vec{v}_i$'s can be written as a linear combination of the others. Again, we may assume that this is true for $\vec{v}_1$, so:

$$\vec{v}_1 = \beta_2\vec{v}_2 + \cdots + \beta_n\vec{v}_n$$

for some $\beta_2, ..., \beta_n \in \mathbb{F}$. Now moving the right-hand side to the left gives us:

$$\vec{v}_1 - \beta_2\vec{v}_2 - \cdots - \beta_n\vec{v}_n = \vec{0}$$

This means $(\alpha_1, ..., \alpha_n) = (1, -\beta_2, ..., -\beta_n)$ is a non-trivial solution to the equation:

$$\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n = \vec{0}$$

and so we have linear dependence of S. $\square$

Example 4.11

Let us consider $\mathbb{F}^2$ and the set:

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

Observe that:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

which means $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Hence, S is linearly dependent.

Example 4.12

Important: for any vector space V, the set $\{\vec{0}\}$ is actually *linearly dependent*! This is because we have $\alpha\vec{0} = \vec{0}$ for all $\alpha \in \mathbb{F}$. So for example, we can take $\alpha = 1$ and we have found a *non-zero* scalar α satisfying $\alpha\vec{0} = \vec{0}$.

Exercise 4.13

Determine which of the following sets are linearly independent or linearly dependent in the given vector spaces. Justify your answers.

(a) $\left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \end{bmatrix} \right\}$ in $\mathbb{R}^2$.

(b) $\{1, 1+x, 1-x\}$ in $\mathcal{P}(\mathbb{F})$.

(c) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$ in $\mathcal{M}_{2\times 2}(\mathbb{R})$.

Lecture 5 Bases and dimension

This covers part of ***Subsection 1.3.1*** *and part of* ***Subsection 1.3.2*** *in the course notes.*

When we study $\mathbb{R}^n$, we work with n coordinates $x_1, ..., x_n$. Each coordinate gives us a different direction we can go in. We call n the *dimension* of $\mathbb{R}^n$ – it tells us precisely how many distinct directions are needed to move in to reach every point in the space.

We can formalize the notion of dimension for abstract vector spaces, but in order to do this we need to define precisely what we should mean by "coordinates". This is done by combining the two notions we looked at in the last lecture: *span* and *independence*.

We fix a vector space V over $\mathbb{F}$. The first lemma below tells us that if we wish to take the span of any finite set S of vectors in V, we can "throw away" any redundant vectors and still get the same span.

Lemma 5.1

Let $S = \{\vec{u}_1, ..., \vec{u}_k\}$ be any subset of V. Then there exists a linearly-independent subset T of S for which $\text{Span}(T) = \text{Span}(S)$.

Proof. If S is linearly independent, we are done. Otherwise, S is linearly dependent, and one of the $\vec{u}_i$'s can be written as a linear combination of the others. By rearranging if necessary, we can assume it is $\vec{u}_k$. Then $S_1 = \{\vec{u}_1, ..., \vec{u}_{k-1}\}$ satisfies $\text{Span}(S_1) = \text{Span}(S)$.

Now if S_1 is linearly independent, we are done. Otherwise, apply the same argument to some other vector in S_1 (assume again it is the last vector $\vec{u}_{k-1}$) to form $S_2 = \{\vec{u}_1, ..., \vec{u}_{k-2}\}$ satisfying $\text{Span}(S_2) = \text{Span}(S_1) = \text{Span}(S)$.

We can repeat this procedure until we reach a point in which we have a linearly-independent subset $T \subseteq S$ satisfying $\text{Span}(T) = \text{Span}(S)$ (which is possible because S is a finite set – we can only throw out so many elements and at the very least the empty set is independent). Thus, T is our desired set. □

Recall that a set $S = \{\vec{v}_1, ..., \vec{v}_n\} \subseteq V$ is a *spanning set* for V when $\text{Span}(S) = V$. The next lemma tells us that if we have a spanning set S for V of size n, any linearly-independent set of vectors in V can have at most size n.

Lemma 5.2

Let $S = \{\vec{v}_1, ..., \vec{v}_n\}$ be a spanning set for V (that is, $\text{Span}(S) = V$) and let $T = \{\vec{w}_1, ..., \vec{w}_k\}$ be a linearly independent set. Then we must have $k \leq n$.

Proof. We will prove this using a familiar tool from MATH 136: solving systems of linear equations. We will prove the contrapositive: if $k > n$, then T must be linearly dependent.

So let us assume we have $k > n$. Since S is a spanning set, for each $1 \leq i \leq k$, we can find coefficients $\alpha_{i,1}, ..., \alpha_{i,n} \in \mathbb{F}$ for which:

$$\vec{w}_i = \alpha_{i,1}\vec{v}_1 + \cdots + \alpha_{i,n}\vec{v}_n$$

Now let $x_1, ..., x_k$ be k unknowns and consider the equation:

$$x_1\vec{w}_1 + \cdots + x_k\vec{w}_k = \vec{0}$$

This can be written and rearranged as follows:

$$\begin{aligned} x_1\vec{w}_1 + \cdots + x_k\vec{w}_k &= \sum_{i=1}^{k} x_i\vec{w}_i \\ &= \sum_{i=1}^{k} x_i(\alpha_{i,1}\vec{v}_1 + \cdots + \alpha_{i,n}\vec{v}_n) \\ &= \sum_{i=1}^{k} x_i \left(\sum_{j=1}^{n} \alpha_{i,j}\vec{v}_j \right) \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{k} \alpha_{i,j}x_i \right) \vec{v}_j \\ &= \vec{0} \end{aligned}$$

Now let us consider the system of n equations in k unknowns $x_1, ..., x_k$:

$$\sum_{i=1}^{k} \alpha_{i,j}x_i = 0 \quad \text{for all } 1 \leq j \leq n$$

Since $k > n$, we know that there must be a non-trivial solution. That is, there exist $\beta_1, ..., \beta_n \in \mathbb{F}$ not all zero such that $(x_1, ..., x_n) = (\beta_1, ..., \beta_n)$ is a solution to the system of equations. In particular, this means:

$$\beta_1\vec{w}_1 + \cdots + \beta_k\vec{w}_k = \vec{0}$$

where not all β_j's are zero. This makes T linearly dependent. □

We are now in a position to combine the two concepts of spanning and independence in the following:

Definition 5.3 (Basis for a vector space)

Let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a subset of V. We say $\mathcal{B}$ is a **basis** for V when:

- $\mathcal{B}$ is a spanning set for V (that is, $\text{Span}(\mathcal{B}) = V$), and
- $\mathcal{B}$ is a linearly-independent set.

There are a couple of strong properties about bases, which we state below.

Theorem 5.4

Let V be a vector space and suppose V has a finite spanning set (that is, there exists a subset $S = \{\vec{u}_1, ..., \vec{u}_k\}$ of V such that $\text{Span}(S) = V$). Then:

(i) V has a basis.

(ii) If $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_m\}$ and $\mathcal{C} = \{\vec{w}_1, ..., \vec{w}_n\}$ are any two bases for V, then we must have $m = n$.

Proof. By **Lemma 5.1**, we can find a linearly-independent subset $\mathcal{B} \subseteq S$ for which $\text{Span}(\mathcal{B}) = \text{Span}(S) = V$. Thus, $\mathcal{B}$ is a basis.

Now let us suppose there are two bases $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_m\}$ and $\mathcal{C} = \{\vec{w}_1, ..., \vec{w}_n\}$ for V. Since $\mathcal{B}$ is spanning and $\mathcal{C}$ is independent, we must have $n \leq m$ by **Lemma 5.2**. But on the other hand, $\mathcal{C}$ is spanning and $\mathcal{B}$ is independent too, so we must also have $m \leq n$, hence $m = n$. □

Definition 5.5 (Finite-dimensional and infinite-dimensional)

We say V is **finite-dimensional** when there exists a *finite* basis $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ for V. Otherwise, we say V is **infinite-dimensional**.

Thanks to the fact that any two bases for a finite-dimensional V must have the same size, it makes sense to give a name to this common size:

Definition 5.6

The **dimension** of a vector space V, denoted $\dim(V)$, is given as follows:

- if V is finite-dimensional, then $\dim(V) = n$ where $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ is any basis for V; and
- if V is infinite-dimensional, then $\dim(V) = \infty$.

Remark 5.7

A basis is what describes a "coordinate system" for a finite-dimensional vector space V. If $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ is a basis for V, we can think of $\vec{v}_1, ..., \vec{v}_n$ as our n coordinates allowing us to reach all the elements of V (the spanning property) in such a way that $\vec{v}_1, ..., \vec{v}_n$ point in mutually different directions (the independence property) so there is no redundancy.

The notion of dimension offers rigidity to our view of a basis being a coordinate system, telling us that if $\dim(V) = n$, then any coordinate system for V must comprise of exactly n coordinates.

Example 5.8 (Standard bases and dimensions)

Many of the spaces we have seen so far are finite-dimensional. They also have what we call *standard bases*. We take a look at them and use them to compute

the dimensions of the corresponding spaces.

(1) In $\mathbb{F}^n$, we have:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad ..., \quad \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

where $\vec{e}_i$ has 1 in the ith position and 0 elsewhere.

The standard basis for $\mathbb{F}^n$ is the set $\{\vec{e}_1, ..., \vec{e}_n\}$. Since this set has n elements, we have $\dim(\mathbb{F}^n) = n$, as we would intuitively expect.

(2) The standard basis for $\mathcal{P}_n(\mathbb{F})$ is the set $\{1, x, x^2, ..., x^n\}$. Thus, $\dim(\mathcal{P}_n(\mathbb{F})) = n + 1$.

(**Small warning:** don't forget that $\mathcal{P}_n(\mathbb{F})$ has dimension $n + 1$, despite having n as a subscript. It is easy to make the mistake of saying $\dim(\mathcal{P}_n(\mathbb{F})) = n$.)

(3) Consider the space $\mathcal{M}_{m\times n}(\mathbb{F})$. For each $1 \leq i \leq m$ and $1 \leq j \leq n$, define $E_{i,j} \in \mathcal{M}_{m\times n}(\mathbb{F})$ as the matrix with 1 in the $(i, j)^{\text{th}}$ position and 0 elsewhere.

The standard basis for $\mathcal{M}_{m\times n}(\mathbb{F})$ is the set $\{E_{i,j} : 1 \leq i \leq m, 1 \leq j \leq n\}$. For example, in $\mathcal{M}_{2\times 3}(\mathbb{F})$, the standard basis is:

$$\{E_{1,1}, E_{1,2}, E_{1,3}, E_{2,1}, E_{2,2}, E_{2,3}\}$$

where:

$$E_{1,1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad E_{1,2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad E_{1,3} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$E_{2,1} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad E_{2,2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad E_{2,3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It follows that $\dim(\mathcal{M}_{m\times n}(\mathbb{F})) = mn$. In particular, $\dim(\mathcal{M}_{2\times 3}(\mathbb{F})) = 6$.

Example 5.9

This example is of the zero vector space $\{\vec{0}\}$, but is important. Notice that $\{\vec{0}\} = \text{Span}(\emptyset)$ and $\emptyset$ is (vacuously) linearly independent. Hence, $\emptyset$ is a basis for $\{\vec{0}\}$ (in fact, this is the *only* basis for it). It follows that $\dim(\{\vec{0}\}) = 0$.

On the other hand, if V is any vector space with $\dim(V) = 0$, this means V has $\emptyset$ as a basis, which forces $V = \{\vec{0}\}$.

Hence, any vector space V is the zero space if and only if it has dimension 0.

Exercise 5.10

Show that $\{1, 1+x, 1+x+x^2\}$ is a basis for $\mathcal{P}_2(\mathbb{F})$.

We will talk more about dimension in the next lecture.

Practice Problems

Here are some practice problems covering what we have done this week.

P1 Let V be a vector space over $\mathbb{F}$.

(a) Prove that if W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is also a subspace of V.

(b) Use part **(a)** and induction to show that if $W_1, W_2, ..., W_n$ are subspaces of V, then $W_1 \cap W_2 \cap \cdots \cap W_n$ is also a subspace of V.

(c) Suppose W_1 and W_2 are subspaces of V. Show that $W_1 \cup W_2$ is a subspace of V if and only if either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

(Hint: for the forward direction, try contradiction. Notice that if W_1 is not a subset of W_2, then there exists a $\vec{w}_1 \in W_1$ with $\vec{w}_1 \notin W_2$, and similarly if W_2 is not a subset of W_1.)

P2 Consider the vector space $\mathcal{P}_3(\mathbb{R})$ over $\mathbb{R}$ and let:

$$U = \{p(x) \in \mathcal{P}_3(\mathbb{F}) : p'(0) = 0\}$$

$$W = \{p(x) \in \mathcal{P}_3(\mathbb{F}) : p'(0) = 1\}$$

where $p'(x)$ denotes the derivative of $p(x)$. For example, if $p(x) = 2x^3 - 3x + 6$ then $p'(x) = 6x^2 - 3$.

(a) Show that U is a subspace of $\mathcal{P}_3(\mathbb{R})$ and find a basis for it.

(b) Determine if W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

P3 Let V be a vector space over $\mathbb{F}$ and let $S_1 = \{\vec{v}_1, ..., \vec{v}_n\}$ and $S_2 = \{\vec{w}_1, ..., \vec{w}_m\}$ be subsets of V.

(a) Prove that if $S_1 \subseteq S_2$ then $\text{Span}(S_1) \subseteq \text{Span}(S_2)$.

(b) Prove that $\text{Span}(S_1 \cap S_2) \subseteq \text{Span}(S_1) \cap \text{Span}(S_2)$.

P4 Consider the vector space $\mathcal{M}_2(\mathbb{F})$ of 2×2 matrices with entries in $\mathbb{F}$. Let:

$$S = \left\{ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \right\}$$

(a) Explain why S *cannot* be a spanning set for $\mathcal{M}_2(\mathbb{F})$.

(b) Determine whether or not S is linearly independent.

P5 Let V be a vector space over $\mathbb{F}$ and let $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ be a set of four vectors in V. Consider the set:

$$T = \{\vec{v}_1 - \vec{v}_2, \vec{v}_2 - \vec{v}_3, \vec{v}_3 - \vec{v}_4, \vec{v}_4\}$$

(a) Show that if S is a spanning set for V, then T is also a spanning set for V.

(b) Show that if S is linearly independent, then T is also linearly independent.

(c) Define:

$$\vec{w}_1 = \vec{v}_1, \quad \vec{w}_2 = \vec{v}_1 + \vec{v}_2, \quad \vec{w}_3 = \vec{v}_1 + \vec{v}_2 + \vec{v}_3, \quad \vec{w}_4 = \vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \vec{v}_4$$

Show that $\text{Span}(S) = \text{Span}(\{\vec{w}_1, \vec{w}_2, \vec{w}_3, \vec{w}_4\})$.

P6 Consider the vector space $\mathbb{C}$ over $\mathbb{F}$ and the set $B = \{1 + \boldsymbol{i}, 1 - \boldsymbol{i}\} \subseteq \mathbb{C}$.

(a) Show that B is linearly independent if $\mathbb{F} = \mathbb{R}$.

(b) Show that B is linearly dependent if $\mathbb{F} = \mathbb{C}$.

P7 Let $x \in \mathbb{R}$ be given and let $S = \{(x, 1, 1), (1, x, 1), (1, 1, x)\} \subseteq \mathbb{R}^3$.

(a) For which values of x is S a linearly-*dependent* subset of $\mathbb{R}^3$?

(b) For each such value of x, compute the dimension of the corresponding subspace spanned by S.